

Data Quality & Forensic Report

Collections Recovery Performance Investigation

Purpose: identify data issues that can distort collections reporting, document the treatment applied, and quantify the business impact before performance and investment decisions.

Executive data-quality assessment: The source environment contains confirmed duplicate, identity and attribution issues. The Golden layer corrects confirmed duplicate-event problems while preserving unresolved identity issues as explicit analytical limitations.

1. Scope & Method

The investigation examined the operational collections datasets for duplicate records, missing values, referential integrity, inconsistent identifiers, payment duplication, borrower/agent identity conflicts, and call/attempt key integrity. Records were inspected before any deletion decision so that legitimate history, retries and source-system collisions were not mistaken for exact duplicates.

2. Data Quality Executive Summary

Issue	Observed result	Treatment	Business impact
Payment duplicates	25,500 raw → 25,000 clean; 500 rows removed	Remove confirmed duplicate copies	Reduces risk of overstated recovery
Missing payment references	382 / 25,000 = 1.53%	Retain NULL; no reliable imputation	Reference-based reconciliation incomplete for 382 records
Borrower ID duplication	8,566 IDs; 19,585 excess records	Do not auto-delete	Borrower-level identity/history remains ambiguous
Agent ID duplication	1,000 IDs; 29,000 excess rows	Do not auto-delete	Person-level agent attribution unreliable
Agent impact	88,241 calls; 28,305 accounts; 15,000 sessions	Flag affected attribution	Agent performance comparisons need identity remediation
Call/attempt integrity	call_id linked to inconsistent operational attributes	Prefer validated attempt-level key	Call-level event attribution can be misleading
WhatsApp duplicates	60,600 source → 60,000 clean; 600 removed	Remove exact duplicates	Prevents digital-event volume inflation; 0 orphan events

3. Payment Data Quality

3.1 Duplicate Payments

Payment IDs were grouped and the underlying duplicate records were inspected before removal. The cleaned payment population fell from 25,500 to 25,000 rows, meaning 500 duplicate rows were removed. The duplicate-payment investigation also identified ₹25,901,961.05 of successful-payment amount associated with removed duplicate records, approximately 1.93% of the unclean successful-payment amount.

Business impact: uncorrected duplicate payment events can inflate transaction counts and reported recovery.

3.2 Missing Payment References

A completeness check found 382 of 25,000 clean payment records with a missing or blank `payment\_reference`, equal to 1.53%. Cross-checking against the Golden recovery population did not produce reliable replacement references.

Treatment: retain NULL values; do not invent or impute references.

Business impact: 382 records remain usable for other payment analysis, but reference-based reconciliation is incomplete.

4. Borrower Identity Integrity

The borrower investigation identified 8,566 borrower IDs appearing multiple times, producing 19,585 excess records. These records are not simple exact duplicates and show variation across borrower attributes.

Treatment: no automatic deletion or 'keep latest row' rule was applied.

Business impact: borrower_id cannot be treated as an unquestionably unique person-level identifier without further identity and temporal-history resolution.

5. Agent Identity Integrity

The agent investigation identified 1,000 duplicated agent IDs and 29,000 excess rows. The same agent ID can correspond to different employee codes, names, vendors, teams, statuses and dates.

Affected population	Count
Calls	88,241
Accounts	28,305
Agent sessions	15,000

Treatment: the agent table was not blindly deduplicated. Aggregate activity remains usable, but person-level agent attribution and causal agent comparisons are treated as unreliable until identity mapping is resolved.

Classification: Fact / Strong Evidence

6. Call / Attempt Integrity

Cross-checks between calls and call_attempts found that the same call_id can be associated with different accounts, borrowers, timestamps, agents and telephony vendors. Therefore, call_id is not assumed to uniquely identify one collection event.

Preferred Golden design: use a validated attempt-level identifier (attempt_id) as the event key and retain call_id as a source/reference field.

7. WhatsApp Data Quality

The WhatsApp source contained 60,600 events. After exact duplicate handling, 60,000 trusted events remained. This represents 600 removed rows, or 0.99% of the source population. No orphan WhatsApp events remained, and no duplicate WhatsApp event IDs remained in the clean population.

8. Source-of-Truth & Treatment Principles

- Do not delete duplicate-looking records until record-level inspection distinguishes exact duplicates from legitimate history or retries.
- Use payment_id as the operational payment identifier after confirmed duplicate handling; keep payment_reference as a descriptive/reconciliation field.
- Attribute borrower through payment.account_id → accounts.account_id → accounts.borrower_id.
- Do not automatically collapse borrower records to the latest row when identities conflict.
- Do not treat agent_id or employee_code as a unique person-level identifier when identity collisions exist.
- Do not assume call_id is a unique event key when call/attempt consistency checks fail.
- Retain unresolved missing references as NULL rather than creating unsupported values.

9. Business Impact on Recovery Reporting

The most material confirmed reporting risk is payment duplication: 500 rows were removed and approximately ₹25.90M of successful-payment amount was associated with removed duplicates. The identity findings also mean that agent- and borrower-level attribution should not be presented as fully reliable without remediation. The cleaned and Golden layers therefore provide a safer foundation for recovery reporting than the raw operational tables.

10. Overall Conclusion

The data contains genuine quality and identity problems that can affect collections reporting. Confirmed duplicate event problems were corrected, while ambiguous borrower and agent identities were preserved and explicitly flagged rather than being 'cleaned' through unsupported assumptions. This approach protects analytical integrity and provides a defensible basis for independently testing the reported 11% month-on-month recovery claim.

11. Recommended Controls for Production

- Primary-key uniqueness and duplicate checks at ingestion.
- Referential-integrity checks between payments, accounts, borrowers and collection events.
- Versioned identity mapping for agents and borrowers.
- Timestamp and timezone validation before hourly/daily operational metrics are calculated.
- Payment reconciliation and duplicate-event monitoring.
- Row-count, status-mix and recovery anomaly alerts.
- Documented metric definitions and lineage from Golden data to Power BI.